

Hien Hoang

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EDUCATION

Worcester Polytechnic Institute, Worcester, MA

May 2026

B.S. in Computer Science (with High Distinction), Minor in Data Science

GPA: 3.8/4.0

Relevant coursework: Distributed Systems, Operating Systems, Database Systems, Computer Networks, Artificial Intelligence, Software Engineering, Algorithms, Object-Oriented Design/Programming, Linear Algebra, Statistics, Probability

Honors & Awards: Best Software Prize @ GoatHacks 2025, 5x Dean's List, WPI Presidential Scholarship, Global Scholarship

EXPERIENCE

Arista Networks

May – August 2026

Software Engineer Intern – Network Detection & Response (Arista NDR)

Austin, TX

- Designed and implemented a **distributed telemetry service in Haskell** on **NixOS/Linux network security appliances** to publish health and configuration data to **Kafka** to enable automated misconfiguration detection and remediation, and capacity-aware bandwidth allocation across heterogeneous sensors.
- Engineered a **least-privilege authorization model** for an internal cloud-based network appliance management and SSH access portal, separating read and connect permissions through **Amazon Cognito** and hierarchical customer/appliance policies in **DynamoDB**.
- Migrated **6 AWS API Gateway endpoints** from direct **DynamoDB** integrations to **AWS Lambda**, enabling fine-grained **Cognito authorization**, decoupling APIs from database layer, and supporting future **PostgreSQL** and **HTTP v2 API** migrations.

DEKA Research & Development

June – December 2025

Cloud Platform Engineer Intern

Manchester, NH

- Designed/implemented **Go REST APIs** for routing **IoT events** and **AWS SSM updates** to downstream microservices.
- Designed/tested **SQLite** data layer, supported **DB selection decision**.
- Cut documentation time by **90% per release** by building **automated documentation CLI tools in Python** for internal release workflows, automatically gathering software version info from GitLab, Artifactory, Jira.
- Integrated tools into **Jenkins CI**, with **95%+ unit test coverage**.

FPT Information Systems

May – August 2024

Software Engineer Intern

Hanoi, Vietnam

- Worked in a team of **4** to build a full-stack HR onboarding platform supporting **200+** new hires, delivering **8+ RESTful APIs** and **5 frontend views** using **React** and **Spring Boot**.
- Improved API response times by **40%** for employee and task endpoints by optimizing **PostgreSQL** queries and implementing **Redis caching** for frequently accessed records.

TECHNICAL SKILLS

Languages: Java, Python, C/C++, Go, Haskell, JavaScript, TypeScript, SQL, HTML, CSS, Bash/Shell

Frameworks & Libraries: React, Express, Spring Boot, FastAPI, Node.js, Socket.io, JUnit, Pytest

Cloud, Databases, Distributed Systems: AWS (Lambda, API Gateway, DynamoDB, EC2, ElastiCache, SQS, SNS, EFS, S3, Cognito), PostgreSQL, MongoDB, Redis, Apache Kafka, Protobuf

Developer Tools: Linux, Docker, Kubernetes, Git, GitHub/GitLab, Jenkins, NixOS, Postman, Confluence, Jira, Figma

PROJECTS

Distributed MapReduce Framework | *Go, RPC, Concurrency*

- Implemented a simplified **MapReduce system in Go** with a coordinator-worker model using **RPC**.
- Built concurrent task scheduling with **goroutines** and **channels**, enabling parallel map/reduce execution while detecting **worker failures** and automatically **reassigning unfinished tasks**.
- Reduced the impact of slow workers by implementing **speculative execution**, launching duplicate copies of **straggling tasks** and accepting the first completed result to improve overall job completion time.

Repo2Resume - WPI 2025 Hackathon Winner | *React, FastAPI, PineconeDB, Docker*

- Built an **AI-powered resume generator** that converts GitHub repositories into STAR-format resume bullets.
- Developed a full-stack system with a **React/Tailwind** frontend and a **FastAPI** backend, containerized with **Docker**.
- Used **PineconeDB** vector database to store job embeddings, enabling **semantic search** to retrieve the most relevant job descriptions and tailor generated resume bullets to target roles.

Database Systems Projects | *Java, File I/O, Data Structures*

- Implemented a **DBMS buffer manager** with frame allocation, Least-Recently-Used (LRU) eviction, and pin/unpin semantics to simulate disk I/O.
- Built indexing structures that reduced query lookup time from **O(n)** scans to **O(1)** access on large datasets.
- Developed a **SQL-like query engine** that selects between table scans and index scans based on query characteristics.